

LangChain overview

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Build completely custom agents and applications powered by LLMs in under 10 lines of code, with integrations for [OpenAI, Anthropic, Google, and more](#). LangChain provides a prebuilt agent architecture and model integrations to help you get started quickly and seamlessly incorporate LLMs into your agents and applications.

LangChain vs. LangGraph vs. Deep Agents

Start with [Deep Agents](#) for a “batteries-included” agent with features like automatic context compression, a virtual filesystem, and subagent-spawning. Deep Agents are built on LangChain [agents](#) which you can also use LangChain directly.

Use [LangGraph](#), our low-level orchestration framework, for advanced needs combining deterministic and agentic workflows.

Use [LangSmith](#) to trace, debug, and evaluate agents built with any of these frameworks. Follow the [tracing quickstart](#) to get set up.

 The [LangSmith Engine](#) detects issues in your LangChain agent traces and proposes fixes. You can open a pull request with the proposed fix directly from the Engine tab.

Create an agent

```
def get_weather(city: str) -> str:
    """Get weather for a given city."""
    return f"It's always sunny in {city}!"

agent = create_agent(
    model="openai:gpt-5.4",
    tools=[get_weather],
    system_prompt="You are a helpful assistant",
)

result = agent.invoke(
    {"messages": [{"role": "user", "content": "What's the weather in San Francisco"}]}
)
print(result["messages"][-1].content_blocks)
```

See the [Installation instructions](#) and [Quickstart guide](#) to get started building your own agents and applications with LangChain.



Use [LangSmith](#) to trace requests, debug agent behavior, and evaluate outputs. Set `LANGSMITH_TRACING=true` and your API key to get started.

Core benefits

Standard model interface

Different providers have unique APIs for interacting with models, including the format of responses. LangChain standardizes how you interact with

Easy to use, highly flexible agent

LangChain's agent abstraction is designed to be easy to get started with, letting you build a simple agent in under 10 lines of code. But it also

[Learn more >](#)

Built on top of LangGraph

LangChain's agents are built on top of LangGraph. This allows us to take advantage of LangGraph's durable execution, human-in-the-loop support, persistence, and more.

[Learn more >](#)

Debug with LangSmith

Gain deep visibility into complex agent behavior with visualization tools that trace execution paths, capture state transitions, and provide detailed runtime metrics.

[Learn more >](#)

 [Connect these docs](#) to Claude, VSCode, and more via MCP for real-time answers.

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